

2026 E Math Prelims P2 Solutions

Qn	Solution
1(a)	$\frac{19a^6b^0c^2}{35c^5} \div \frac{9a^2b^4}{14c}$ $= \frac{19a^6c^2}{35c^5} \times \frac{14c}{9a^2b^4}$ $= \frac{38a^4}{45b^4c^2}$
1(b)	$\left(\frac{12x^{-4}y^6}{27} \right)^{\frac{1}{2}}$ $= \left(\frac{27}{12x^{-4}y^6} \right)^{\frac{1}{2}}$ $= \left(\frac{9x^4}{4y^6} \right)^{\frac{1}{2}}$ $= \frac{3x^2}{2y^3}$
1(c)	$2^{x+1} = 2^x + 32$ $2(2^x) = 2^x + 32$ $2^x = 32$ $x = 5$
1(d)	$\frac{x+2}{x-3} - \frac{60}{2x^2-18}$ $= \frac{x+2}{x-3} - \frac{60}{2(x^2-9)}$ $= \frac{x+2}{x-3} - \frac{30}{(x^2-9)}$ $= \frac{(x+2)}{x-3} - \frac{30}{(x-3)(x+3)}$ $= \frac{(x+2)(x+3)-30}{(x-3)(x+3)}$ $= \frac{x^2+5x-24}{(x-3)(x+3)}$ $= \frac{(x+8)(x-3)}{(x-3)(x+3)}$ $= \frac{x+8}{x+3}$
(e)	The numbers $n(n^2-1) = n(n-1)(n+1)$ are consecutive integers. Among any two consecutive integers, one is even. Therefore, at least one of the factors is even in $n(n^2-1)$. Hence, $n(n^2-1) = n(n-1)(n+1)$ is always even. Alternatively,

	accept if students explicitly mention that if n is <u>odd</u> /even, then (n-1) and (n+1) is <u>even</u> /odd. Therefore $n(n^2 - 1) = n(n-1)(n+1)$ is always even.
2(a)	$(x - p)(x + 2p) = 0$ $x^2 + 2px - px - 2p^2 = 0$ $x^2 + px - 2p^2 = 0$
(b)	$\frac{(2 + \sqrt{a}) + (2 - \sqrt{a})}{2}$ $= 2$ $x = 2$
(ci)	$\vec{PQ}$ $= \begin{pmatrix} -9 \\ 3 \end{pmatrix} - \begin{pmatrix} -5 \\ 9 \end{pmatrix}$ $= \begin{pmatrix} -4 \\ -6 \end{pmatrix}$
(cii)	Accept any answer in the form $k \begin{pmatrix} -4 \\ -6 \end{pmatrix}$ Where k is any constant.

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3(ai)	$\mathbf{P} = \begin{pmatrix} 80 \\ 30 \end{pmatrix}$
(ii)	$\mathbf{T}$ $= \mathbf{WP}$ $= \begin{pmatrix} 12 & 4 \\ 9 & 5.6 \\ 11.8 & 3 \end{pmatrix} \begin{pmatrix} 80 \\ 30 \end{pmatrix}$ $= \begin{pmatrix} 1080 \\ 888 \\ 1034 \end{pmatrix}$
(iii)	The elements represent the <u>total merchandise revenue, in thousands of dollars, for each game respectively.</u> Or \$1080000, \$888000, \$1034000 are the total revenue collected from game 1,2 and 3 respectively.
(b)	$d \begin{pmatrix} 7 & 8 \\ 0 & 30 \end{pmatrix} \begin{pmatrix} 80 & 0 \\ 0 & 30 \end{pmatrix} = \begin{pmatrix} 448 & 192 \end{pmatrix}$ $(7d \quad 8d) \begin{pmatrix} 80 & 0 \\ 0 & 30 \end{pmatrix} = \begin{pmatrix} 448 & 192 \end{pmatrix}$ $(560d \quad 240d) = \begin{pmatrix} 448 & 192 \end{pmatrix}$ $560d = 448$ $d = 0.8$ <i>Discount</i> $= 1 - 0.8$ $= 0.2$ Hence the percentage discount given is 20%

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4(ai)	$y = \frac{1}{2}(-1)^3 - \frac{1}{2}(-1)^2 - 2(-1) + 2$ $y = -\frac{1}{2} - \frac{1}{2} + 2 + 2$ $y = 3$
4(aii) (iii)	
(iv)	$X = 0, 2$
(v)	$x > 2$
(b)	$y = ka^x$ $4 = ka^0$ $k = 4$ $a = 2$
(5)(a)	$AO = BO(\text{radius})$ $AP = BP(\text{tangents from external point})$ PO is common side Hence $\triangle AOP \equiv \triangle BOP$ (SSS)
5(b)	2 radians Angle at center = twice angle at circumference

5(c)

$$\angle AOP = 1 \text{ rad}$$

Since $\triangle OAP$ is a right angled triangle,

$$\tan \angle AOP = \frac{AP}{AO}$$

$$\tan(1 \text{ rad}) = \frac{AP}{6}$$

$$AP = 6 \tan(1 \text{ rad})$$

$$AP = 9.344446348$$

$$\text{Area } \triangle OAP = \frac{1}{2} \times 6 \times 9.344446348$$

$$\text{Area } \triangle OAP = 28.03333904 \text{ cm}^2$$

Area of Sector AOQ

$$= \frac{1}{2} (6)^2 (1)$$

$$= 18 \text{ cm}^2$$

Shaded region APQ

$$= \text{Area } \triangle OAP - \text{Area of Sector AOQ}$$

$$= 28.03333904 - 18$$

$$= 10.0 \text{ cm}^2 (3 \text{ sf})$$

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6(ai)	$33 = \frac{1}{2}(7)(12)\sin BAC$ $\sin BAC = \frac{33}{42}$ $BAC = \sin^{-1}\left(\frac{33}{42}\right)$ $BAC = 51.7867893$ $BAC = 51.8^\circ (3sf)$
(ii)	$BC^2 = 7^2 + 12^2 - 2(7)(12)\cos 51.8$ $BC = \sqrt{7^2 + 12^2 - 2(7)(12)\cos 51.8}$ $BC = 9.439671053cm$ $BC = 9.44cm(3sf)$
(iii)	$\frac{\sin \angle BAC}{BC} = \frac{\sin \angle ACB}{AB}$ $\frac{\sin 51.8}{9.439671053} = \frac{\sin \angle ACB}{7}$ $9.439671053 \sin \angle ACB = 7 \sin(51.8)$ $\angle ACB = 35.64442002^\circ$ $\angle ACB = 35.6^\circ (1dp)$ $\text{Bearing } B \text{ from } C$ $= 270 + 35.6$ $= 305.6^\circ$
(b)i	$Area = 33$ $\frac{1}{2} \times 12 \times height = 33$ $height = \frac{33}{6} = 5.5km$
(bii)	$\tan \theta = \frac{0.06}{5.5}$ $\theta = \tan^{-1}\left(\frac{0.06}{5.5}\right)$ $\theta = 0.6^\circ (1dp)$

7(a)	$\text{Time taken to store} = \frac{4}{x}h$ $\text{Time taken to home} = \frac{4}{x-2.5}h$ $\text{Total time on the road} = 60 - 20 = 40 \text{ min} = \frac{2}{3}h$ $\frac{4}{x} + \frac{4}{x-2.5} = \frac{2}{3}$ $\frac{4(x-2.5) + 4x}{x(x-2.5)} = \frac{2}{3}$ $12(x-2.5) + 12x = 2x(x-2.5)$ $24x - 30 = 2x^2 - 5x$ $2x^2 - 29x + 30 = 0(\text{shown})$
7(b)	$2x^2 - 29x + 30 = 0$ $x = \frac{-(-29) \pm \sqrt{(-29)^2 - 4(2)(30)}}{2(2)}$ $x = 1.121 \text{ or } 13.379 \text{ (3dp)}$
7(c)	Reject 1.121, since $1.121 - 2.5$ would give a negative return speed .
7(d)	$2x^2 - 29x + 30 = 0$ $x = 13.379 \text{ or } 1.121(\text{reject})$ $x - 2.5 = 13.379 - 2.5$ $\text{return speed} = 10.9 \text{ km / h (3sf)}$

8(a)i	$x = 49$
(ii)	Mean $\frac{15560}{150}$ $= 103.733333..$ $= 104 \text{ min s (3sf)}$
(iii)	$\sqrt{\frac{(60^2 \times 12) + (80^2 \times 29) + (100^2 \times 49) + (120^2 \times 39) + (140^2 \times 21)}{150} - \left(\frac{15560}{150}\right)^2}$ $= \sqrt{\frac{1692000}{150} - \left(\frac{15560}{150}\right)^2}$ $= \sqrt{\frac{1692000}{150} - \left(\frac{15560}{150}\right)^2}$ $= 22.8 \text{ min s (3sf)}$
(iv)	99.5 < 103.7: Summit Polytechnic participants were faster on average. 25.6 > 22.9: Regent Polytechnic participants were more consistent. therefore it is not justifiable as it depends on whether average speed or consistency is considered as the criteria for "better" performance across the group.

(bi)	$\frac{27}{100} \times \frac{26}{99}$ $= \frac{39}{550}$
(ii)	$1 - \left(\frac{73}{100} \times \frac{72}{99} \times \frac{71}{98} \right)$ $= \frac{8292}{13475}$
(iii)	Histogram shows grouped data, not the exact finishing . The exact numbers on each side of 5.5 and 6.5 cannot be determined from the histogram.

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9(a)	Area of whole roof = $12 \times 7 = 84$ Area not connected = $3 \times 2 = 6$ Collecting area = $84 - 6 = 78$ Since 1mm of rain on 1m^2 gives 1litre of water, 15 mm of rain on 78 m^2 gives: $78 \times 15 = 1170$ litres Only 82% is collected: Volume collected $= 0.82 \times 1170$ $= 959.4$ litres
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(b)

Water needed for 5 days

$$= 5 \times 170$$

$$= 850 \text{ litres}$$

Since $959.4 \geq 850$, one heavy shower collects enough water to meet this requirement.

Volume of Tank A

$$= \pi r^2 h$$

$$= 3.142 \times 0.5^2 \times 1.1$$

$$= 0.86405 \text{ m}^3$$

$$= 864 \text{ litres (3 s.f.)}$$

With safety feature, volume

$$= 90\% \times 864.05$$

$$= 777.6 < 850$$

$\therefore$ Tank A is not suitable as there are not enough for 5 days without rain

Volume of Tank B

$$= 1.1 \times 1.0 \times 1.1$$

$$= 1.21 \text{ m}^3$$

$$= 1210 \text{ litres}$$

With safety feature, volume

$$90\% \times 1210 = 1089 \geq 850$$

$\therefore$ Tank B is suitable

Volume of Tank C

$$= 3.142 \times 0.6^2 \times 0.9$$

$$= 1.018 \text{ m}^3$$

$$= 1018 \text{ litres (4 s.f.)}$$

With safety feature, volume

$$90\% \times 1018.008 = 916.2 \geq 850$$

$\therefore$ Tank C is suitable

Budget Check

$$\text{Tank B: } \$680 + \$120 = \$800 \leq \$800$$

$$\text{Tank C: } \$710 + \$80 = \$790 \leq \$800$$

Both B and C satisfy the capacity and budget requirements.

Tank C is recommended since it is \$10 cheaper (\$790 vs \$800).